

GARMIN®

AVIATION TRAINING

For Reference Only
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Engine Information Systems

GI-275 EIS both Dealer and
Over the Counter advice

Garmin Aviation Training is intended to provide the best practices for the installation, use and maintenance of Garmin Aviation products. It is incumbent on technicians and pilots to ensure that the instructions contained in the most recent versions of Garmin product manuals, bulletins and notifications are followed.

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Objectives

Monday, March 30, 2026

- ✈ Focus on GI-275 Engine Indication Systems
- ✈ Configuration Walk-Thru
 - ✈ EIS workflow as outlined by Garmin Documents
- ✈ Quiz

Objectives

Monday, March 30, 2026

- ✈ After completing this session, you will:
 - ✈ Understand the configuration workflow for Garmin Engine Indication Systems as they relate to the Garmin Indicator 275.
 - ✈ Better articulate the recipe, either an independent technician or Garmin Dealer, for following Garmin STC Documentation available from Garmin.com and the DRC.

✈️ Best workflow for EIS Configs:

- ✈️ Engine
- ✈️ Sensors
- ✈️ Pages
- ✈️ Gauges
- ✈️ Fuel
- ✈️ Diagnostics



✈️ Best workflow for EIS Configs:

✈️ Identify

✈️ Configure

✈️ Calibrate



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✈ Identify

✈ Engine Parameters

✈ Power Settings

✈ Hours

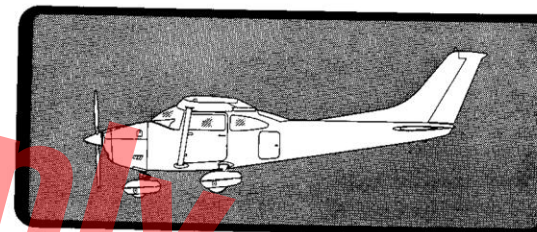
✈ Flight

✈ Hobbs

✈ Tach

PILOT'S OPERATING HANDBOOK

Cessna



SKYLANE

1977 MODEL 182Q

Serial No. _____

Registration No. _____

THIS HANDBOOK INCLUDES THE MATERIAL
REQUIRED TO BE FURNISHED TO THE PILOT
BY CAR PART 3

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CESSNA AIRCRAFT COMPANY
WICHITA, KANSAS, USA

D1087-13-RPC-1000-3/78

✈️ Identify

✈️ Engine Parameters

✈️ Power Settings

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CESSNA
MODEL 182Q

SECTION 1
GENERAL

SECTION 1 GENERAL

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✈ Identify

✈ Engine Parameters

✈ Power Settings

✈ Hours

✈ Flight

✈ Hobbs

✈ Tach

CESSNA
MODEL 182Q

SECTION 1
GENERAL

INTRODUCTION

This handbook contains 9 sections, and includes the material required to be furnished to the pilot by CAR Part 3. It also contains supplemental data supplied by Cessna Aircraft Company.

Section 1 provides basic data and information of general interest. It also contains definitions or explanations of symbols, abbreviations, and terminology commonly used.

DESCRIPTIVE DATA

ENGINE

Number of Engines: 1.
Engine Manufacturer: Teledyne Continental.
Engine Model Number: O-470-U.
Engine Type: Normally-aspirated, direct-drive, air-cooled, horizontally-opposed, carburetor-equipped, six-cylinder engine with 470 cu. in. displacement.
Horsepower Rating and Engine Speed: 230 rated BHP at 2400 RPM.

PROPELLER

Propeller Manufacturer: McCauley Accessory Division.
Propeller Model Number: C2A34C204/90DCB-8.
Number of Blades: 2.
Propeller Diameter, Maximum: 82 inches.
Minimum: 80.5 inches.
Propeller Type: Constant speed and hydraulically actuated, with a low pitch setting of 15.0° and a high pitch setting of 29.4° (30 inch station).

FUEL

Approved Fuel Grades (and Colors):
100LL Grade Aviation Fuel (Blue).
100 (Formerly 100/130) Grade Aviation Fuel (Green).

✈️ Identify

✈️ Engine Parameters

✈️ Power Settings

✈️ Hours

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- ✈️ Sensors

✈️ Configure

- ✈️ Pages
- ✈️ Gauges
- ✈️ Fuel

✈️ Calibrate

- ✈️ Sensors/Gauges
- ✈️ Diagnostics



✈ After Identifying our GI we need to tell it what it's job is in the EFIS or as a stand alone installation.

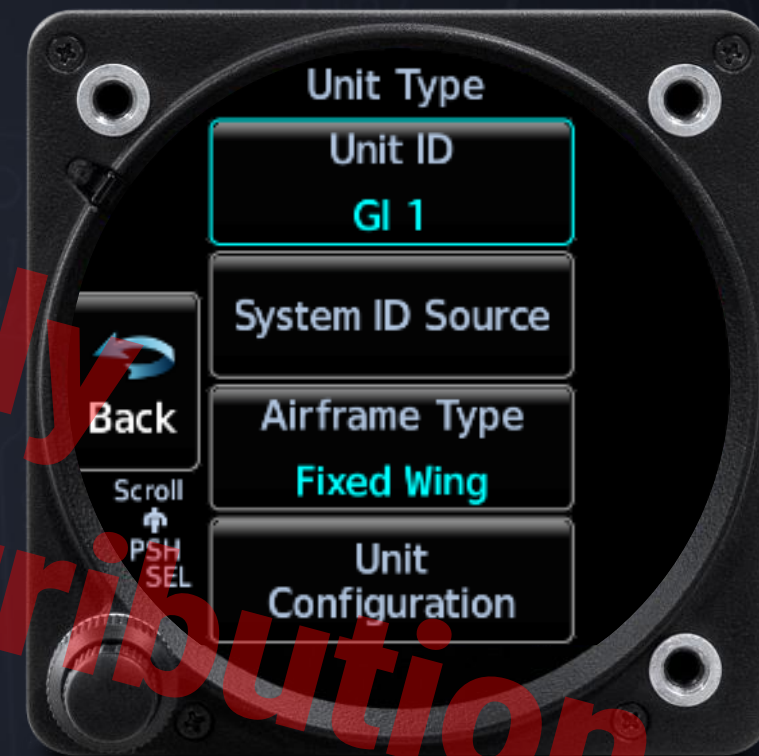
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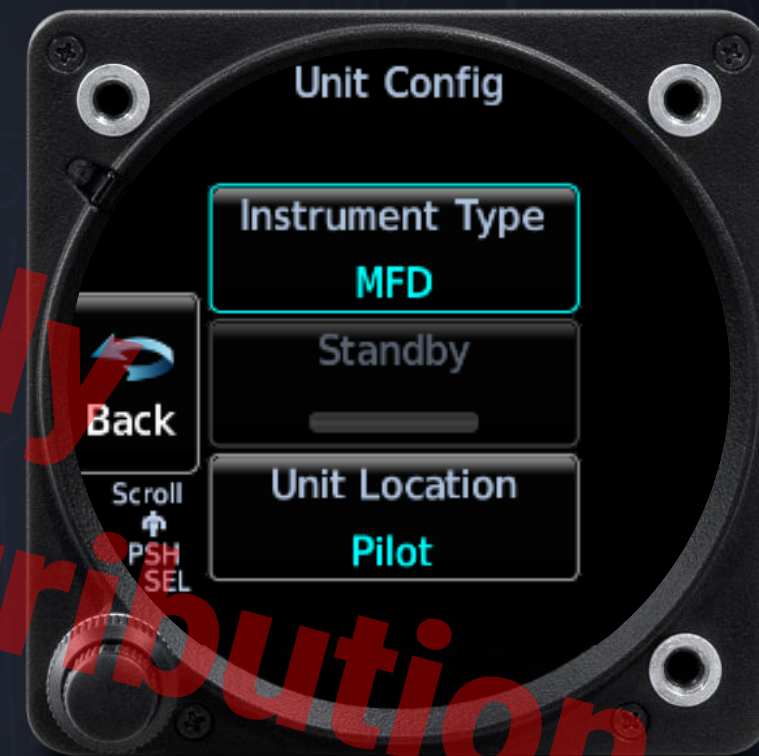


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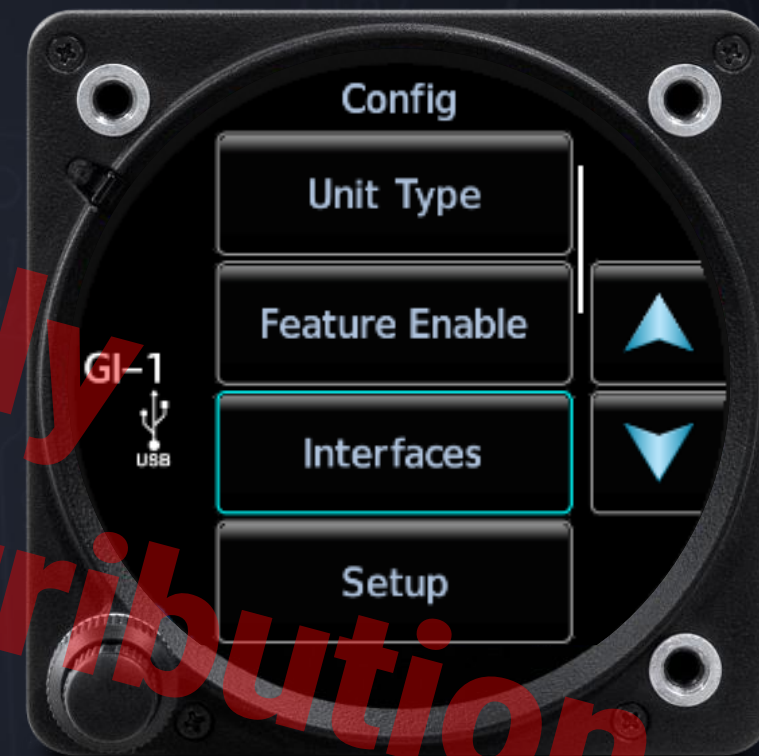
✈️ Then EIS



- ✈️ Now we've told our GI 275 it's doing.
 - ✈️ This is the process regardless if we're setting up a single GI or several in a larger EFIS arrangement.
- ✈️ Next we'll return home so we may go to the 'Interfaces' menu and select our GEA and it's interfaced port.



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✈️ Scroll down to EIS 1

✈️ This represents the EIS for the #1 Engine. So if this were a twin, you might choose to configure EIS 2.



✈ Tap EIS 1 to configure the Garmin Engine Adapter system for the engine you're configuring.

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✈ EIS 1

✈ Tap Interface to choose your GEA

✈ GEA 24

✈ Also tell the GI which RS 232 the GEA 24 is using.

✈ Alternatively, you can set the RS 485 port you're using with your GEA 110.



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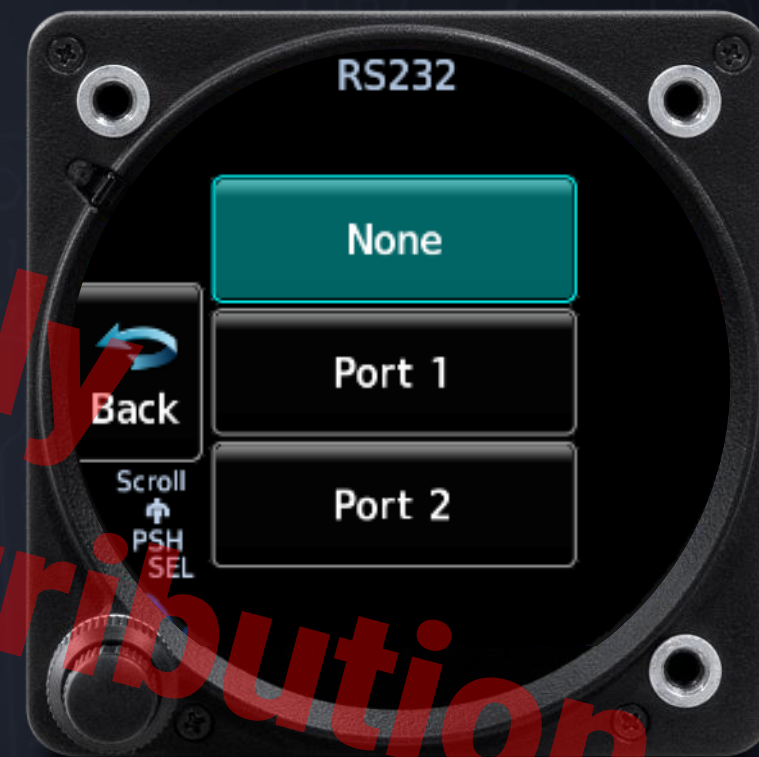
✈ EIS 1

✈ Tap Interface to choose your GEA

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✈ Also tell the GI which RS 232 the GEA 24 is using.

✈ Alternatively, you can set the RS 485 port you're using with your GEA 110.



✈️ Note:

- ✈️ You can route the Warning and Caution lighting discrets from the 'Annunciation' menu below the RS 232 selection.



✈ Best work flow for EIS
Configs:

✈ Configure

✈ Sensors

✈ Pages

✈ Gauges

✈ Fuel



- ✈ Each connected EIS sensor is configured following this process:
 - ✈ Determine all installed EIS sensors that interface to the GI 275 system. (Identify)
 - ✈ From the 'Home' configuration menu navigate to then tap 'EIS'
 - ✈ Once you've input the Engine Parameters:
 - ✈ Cylinder number
 - ✈ Power metrics
 - ✈ Hours



✈ In both the Over the Counter and the Approved Model List STC manuals you'll find this walk thru for a TIT sensor.

✈ Sensor configurations are in Appendix C.

✈ Table C.12

✈ Follow the menu walk thru as follows.



- ✈ Tap the 'EIS' Menu and then tap 'Sensors'
- ✈ Scroll to the sensor you're going to setup.
- ✈ We're doing the TIT example which is in the Install Manual
- ✈ To begin we'll assume you've tapped the 'EIS' menu and then selected 'Sensors'



- ✈️ Scroll down to the TIT selection and tap the window indicating you'd like to configure this sensor.



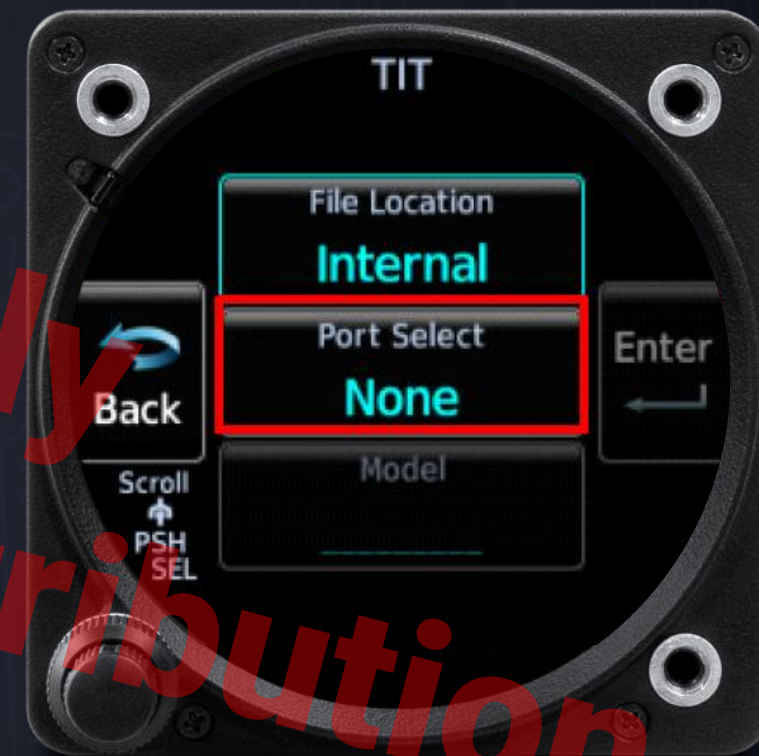
- ✈️ File Location is used to configure from the list of optional sensors.
- ✈️ Tap 'File Location' and choose 'Internal' if the GI will be the origin for this configuration.
 - ✈️ Alternatively, you may select the USB drive if the configuration exists on a connected USB Storage device.



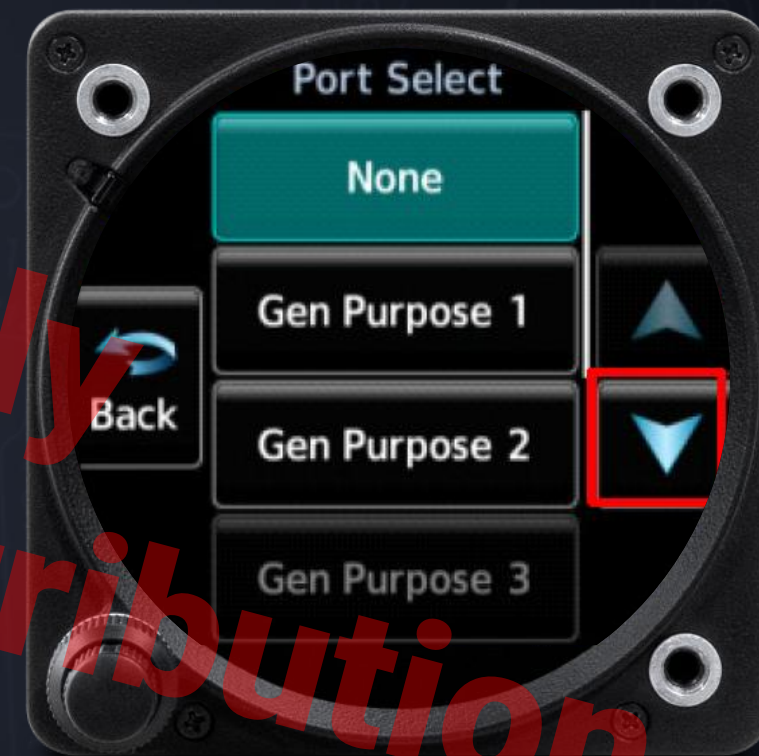
- ✈ We'll choose 'Internal' for this example to match the Installation Manual.



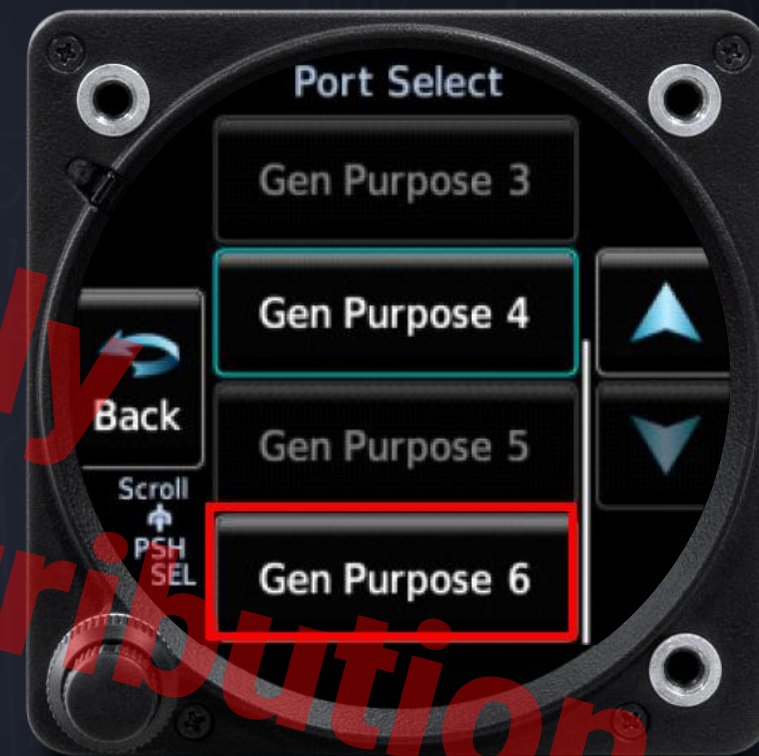
- ✈️ Next we're telling the GI EIS how to communicate with the TIT sensor.
- ✈️ Let's choose the input where the TIT is connected on the GEA.



✈ Scroll down:



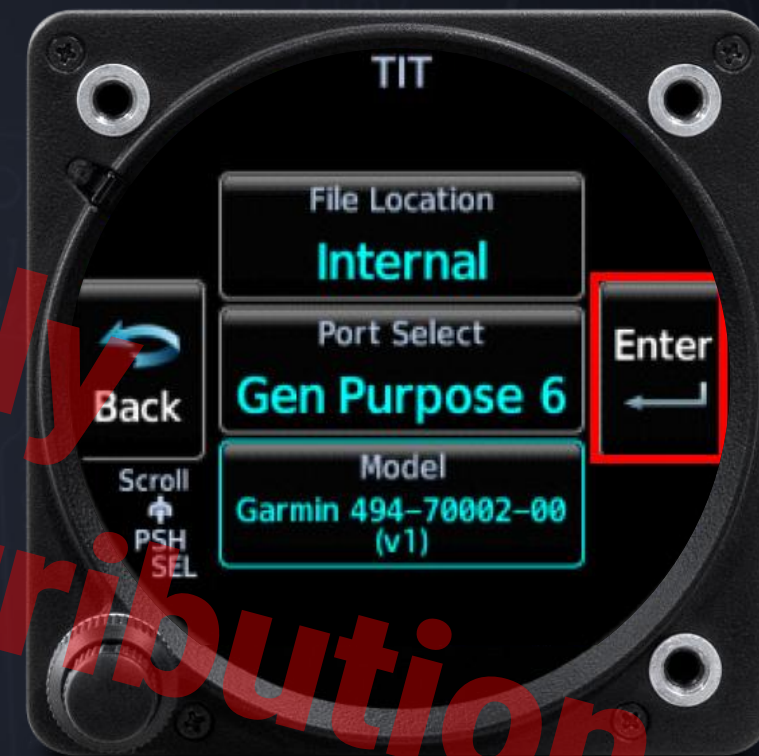
- ✈ In this case the TIT sensor is connected to the 'Gen Purpose 6' input on the GEA 24.



✈️ Next we'll tap the 'Model' menu so we can choose the correct sensor.



- ✈ In this case we've chosen the Garmin Part Number 494-70002-00
 - ✈ This is a standard threaded bayonet style 'K Type' Thermocouple.
 - ✈ Don't forget to tap 'Enter' to confirm instead of 'Back' or you'll find you need to repeat a step or two.



- ✈ You'll walk thru this process for each sensor interfaced to the GEA.
- ✈ To ensure the gauges and pages appear and operate the way we expect we'll follow this basic process outlined in Chapter 5 section 7.2 for each sensor system.
- ✈ When completed we can move on to the pages and gauges.



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✈️ **Best** workflow for EIS
Configs:

✈️ **Calibrate**

✈️ Once the Sensors are setup, time to make them work and report as required



✈️ **Best** workflow for EIS
Configs:

✈️ **Calibrate**

✈️ Once the Sensors are setup, time to make them work and report as required



✈ Quick Tips on EIS Configs:

✈ Improperly setting the K Factor of your flow transducers.

✈ Use manufacturer's settings: Sometimes this means reaching out to the manufacturer with the model and serial numbers to retrieve this information.

✈ Set the K Factor first then set the sensor type.

✈ If K Factor ever needs adjusted, you must reset the sensor to 'None' before changing.



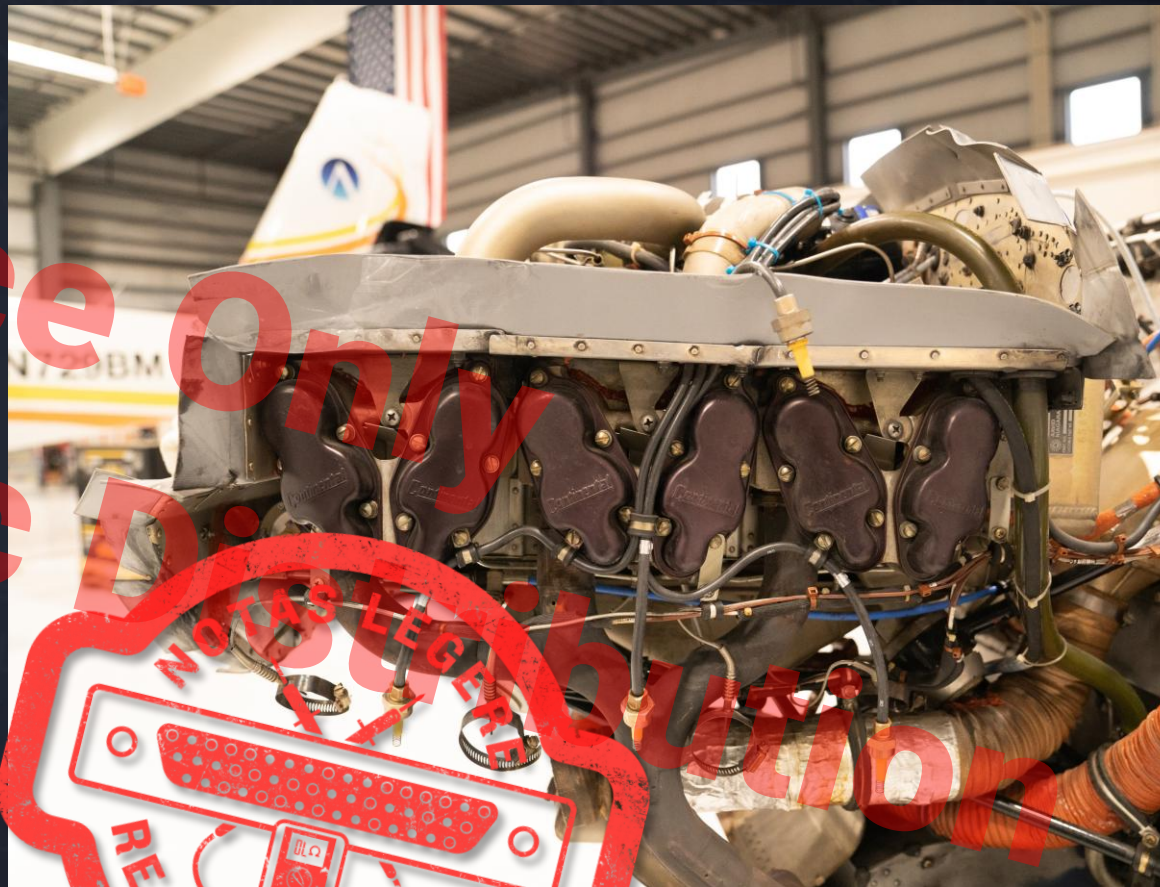
✈ Quick Tips on EIS Configs:

- ✈ Ensure correct Fuel Quantity sensors are configured before calibration
- ✈ Pay attention to Fuel Totals calibrate for individual tank and make sure your aircraft total capacity is correct.
 - ✈ Per Tank QTY = EIS -> GAUGES -> FUEL (Main)
 - ✈ Total Combined QTY = EIS -> GAUGES -> FUEL -> FULL CAPACITY
- ✈ Ensure each configured Gauge is a match for the removed equipment.



✈️ EIS Double Checks:

- ✈️ Once installed, configured, and calibrated a sensor may act up:
 - ✈️ Check splices
 - ✈️ Ensure wire shields and insulators are intact
 - ✈️ Examine bonding of all components in the system
- ✈️ Check your Appendices
- ✈️ NOTAS LEGERE!





What Installation Manual section walks you thru EIS Configuration?

- A. Chapter 5, Section 5
- B. Chapter 5, Section 7
- C. Appendix C, Section 5
- D. Appendix E, Section 1



GI 275 Engine Indication Systems offer which of the following?

- A. 4 Cylinder Piston Engines (GEA 24)
- B. 7 Cylinder Radial Engines (GEA 110)
- C. Turbine Engines (GEA 71B)
- D. 4 Cylinder Diesel Engines (GEA 71B)



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I participated in the Red Room EIS Configuration Section

- A. True
- B. False

Summary

Monday, March 30, 2026

- ✈ We broke down the Config Workflow
 - ✈ Identify
 - ✈ Configure
 - ✈ Calibrate
- ✈ We looked at the Configuration of a sensor
- ✈ Examples we explored are in the OTC and AML STC installation documentation so A&P's and Garmin Dealers alike can install Garmin GI 275 EIS Systems.





Any Questions?

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Thank You!

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